SOLAR HOUSE (CASA SOLARIS)

Why is this project a best practice?s

Challenge

The project had to provide high thermal comfort through cold winters and warm summers while sharply reducing operational energy. The main design challenge was to coordinate a high-performance envelope, efficient services, renewable energy and healthy materials without compromising the architectural concept, construction quality or everyday usability.

Solution

The response combines 72 m² of photovoltaic panels with surplus electricity exported to the public grid; 37 m² of solar-thermal collectors serving domestic hot water and winter heating; and underground seasonal storage of summer heat, avoiding a heat pump, combined with an energy demand reported at about 50 kWh/m²/year. These measures were brought together through an integrated design approach and, where applicable, the Green Homes, nZEB or Passive House performance framework.

Replicability

The core package - demand reduction first, airtight construction, efficient low-temperature systems, ventilation and on-site renewables - is transferable to other Romanian homes. Replication requires climate-specific energy modelling, careful thermal-bridge and airtightness detailing, qualified installation, commissioning and clear operating guidance for residents.

Key facts and figures

Location	Voluntari, Ilfov County, Romania
Building use / function	Single-family dwelling

Project type (new build, refurbishment, extension)	New build; active-house pilot
Plot area / Floor area	Not publicly disclosed; first of three planned pilot homes
Year / phase	Completed and Green Homes-certified in 2014
Climate zone	Temperate continental
Thematic focus / innovation	Active house; 72 m² photovoltaic array; 37 m² solar thermal; seasonal heat storage; low-energy envelope
Investor / building owner	Casa Solaris SRL
Architect	Not publicly disclosed in the consulted sources
Other involved stakeholders	Romania Green Building Council; solar, thermal-storage and smart-control specialists
Contact person	Casa Solaris SRL / RoGBC
Further information	https://www.rogbc.org/proiecte-certificate-rogbc?lang=en https://www.rogbc.org/_files/ugd/40ee3e_b5d91d7a3b71422cb98cd7a289b89c79.pdf

Solar House (Casa Solaris) is a new build; active-house pilot project in Voluntari, Ilfov County, Romania. It functions as single-family dwelling. Its main best-practice contribution is An early Romanian active-house pilot that combines solar electricity, solar heat and seasonal storage. The documented strategy brings together 72 m² of photovoltaic panels with surplus electricity exported to the public grid, 37 m² of solar-thermal collectors serving domestic hot water and winter heating, and underground seasonal storage of summer heat, avoiding a heat pump, combined with an energy demand reported at about 50 kWh/m²/year. The RoGBC toolkit reports that the house produced more energy than its current operation required in 2014. Public-source research was reviewed in July 2026; data that vary by phase are explicitly flagged in this sheet.

What was done differently?

1. Performance concept - 72 m² of photovoltaic panels with surplus electricity exported to the public grid.
2. Integrated systems and operation - 37 m² of solar-thermal collectors serving domestic hot water and winter heating.
3. Wider value - Underground seasonal storage of summer heat, avoiding a heat pump, combined with an energy demand reported at about 50 kwh/m²/year.

What was achieved?

Impact

Environmental impact: The project reduces energy and associated emissions primarily by lowering demand and then using efficient or renewable systems. Its material, landscape, waste or mobility measures add benefits beyond operational energy where documented. A verified whole-life carbon assessment was not publicly available, so the impact is described qualitatively unless a source provides a specific figure.

Economic impact: Lower energy demand can reduce utility exposure and improve long-term operating-cost predictability. Standardised high-performance specifications and third-party certification can also support quality assurance, market value and access to green-finance products. Public sources do not provide a complete investment, payback or life-cycle-cost analysis for this project.

Social impact: The main social benefits are stable indoor temperatures, better air quality, daylight and acoustic comfort, together with safer and more attractive shared or outdoor spaces where these form part of the project. The project also raises market awareness by making high-performance housing visible to residents, designers, contractors and investors.

Key Learnings

Success factors: Key success factors are an integrated design process, a clear performance framework, early coordination of envelope and building services, selection of competent suppliers, quality control during construction and commissioning. Transparent documentation and communication with occupants are essential to preserve the intended performance in use.

Lessons for replication: Replication should start with an energy and comfort target, not a catalogue of technologies. Design teams should verify thermal bridges, airtightness, ventilation, controls and renewable sizing together; record assumptions by project phase; commission systems; and monitor early operation. Where public figures differ by phase, the final as-built dataset must become the single source of truth.